Exam AWS Certified Developer - Associate DVA-C02 All Questions

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EXAM AWS CERTIFIED DEVELOPER - ASSOCIATE DVA-C02 TOPIC 1 QUESTIO...

Exam question from Amazon's AWS Certified Developer - Associate DVA-C02
Question #: 227
Topic #: 1
[All AWS Certified Developer - Associate DVA-C02 Questions]

A developer has built an application that inserts data into an Amazon DynamoDB table. The table is configured to use provisioned capacity. The application is deployed on a burstable nano Amazon EC2 instance. The application logs show that the application has been failing because of a ProvisionedThroughputExceededException error.

Which actions should the developer take to resolve this issue? (Choose two.)

A. Move the application to a larger EC2 instance.

B. Increase the number of read capacity units (RCUs) that are provisioned for the DynamoDB table.

C. Reduce the frequency of requests to DynamoDB by implementing exponential backoff.

D. Increase the frequency of requests to DynamoDB by decreasing the retry delay.

E. Change the capacity mode of the DynamoDB table from provisioned to on-demand.

Show Suggested Answer

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by Claire_KMT at *Oct. 28, 2023, 2:12 p.m.*

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Comments

bhanupriya07 Highly Voted 2 years, 4 months ago

Selected Answer: CE

C. Reduce the frequency of requests to DynamoDB by implementing exponential backoff.
E. Change the capacity mode of the DynamoDB table from provisioned to on-demand.

upvoted 7 times

chikuwan Highly Voted 2 years, 1 month ago

Selected Answer: CE

What the question said is insert data..so B increase read capacity is not correct.Hence C and E.

upvoted 5 times

MasoudK Most Recent 1 year, 5 months ago

B And C: While E could resolve the issue by automatically scaling the capacity, it may not be the most cost-effective solution if the traffic pattern is predictable and can be managed with provisioned capacity and backoff strategies.

upvoted 2 times

albert_kuo 1 year, 6 months ago

Selected Answer: BC

By implementing exponential backoff, the application will automatically retry failed requests with increasing delays between attempts. This helps to spread out the requests over time, reducing the likelihood of exceeding the provisioned throughput.

Since the table is configured to use provisioned capacity, increasing the number of RCUs will allow the table to handle more read requests per second. This directly addresses the ProvisionedThroughputExceededException by increasing the table's capacity to handle more concurrent requests.

upvoted 1 times

ahadh7621 1 year, 8 months ago

Selected Answer: CE

Answer is C & E:
https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/Programming.Errors.html

ProvisionedThroughputExceededException
Message: You exceeded your maximum allowed provisioned throughput for a table or for one or more global secondary indexes. To view performance metrics for provisioned throughput vs. consumed throughput, open the Amazon CloudWatch console.

Example: Your request rate is too high. The AWS SDKs for DynamoDB automatically retry requests that receive this exception. Your request is eventually successful, unless your retry queue is too large to finish. Reduce the frequency of requests using Error retries and exponential backoff.

upvoted 1 times

65703c1 1 year, 10 months ago

Selected Answer: CE

CE is the correct answer.

upvoted 2 times

KarBiswa 2 years ago

Selected Answer: CE

https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/HowItWorks.ReadWriteCapacityMode.html

upvoted 3 times

KarBiswa 2 years, 1 month ago

Selected Answer: BC

https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/ProvisionedThroughput.html#:~:text=The%20DynamoDB%20console%20displays%20Amazon%20CloudWatch%20metrics%20for%20your%20tables%2C%20so%20you%20can%20monitor%20throttled%20read%20requests%20and%20write%20requests.%20If%20you%20encounter%20excessive%20throttling%2C%20you%20should%20consider%20increasing%20your%20table%27s%20provisioned%20throughput%20settings.

upvoted 1 times

KarBiswa 2 years ago

choosing C & E after going through https://docs.aws.amazon.com/amazondynamodb/latest/developerguide/HowItWorks.ReadWriteCapacityMode.html

upvoted 1 times

SerialiDr 2 years, 2 months ago

Selected Answer: BC

B. This error indicates that the application's request rate is exceeding the throughput that has been provisioned for the table. Increasing the provisioned read capacity units (RCUs) and/or write capacity units (WCUs) for the DynamoDB table will allow it to handle a higher request rate, thereby reducing the likelihood of encountering this error. However, this approach requires careful capacity planning and may increase costs.

C. Exponential backoff is a standard error retry strategy that involves progressively increasing the delay between retries when there is a ProvisionedThroughputExceededException. This approach helps to smooth out the rate of requests, giving the table time to accommodate bursts of read or write requests. Implementing exponential backoff in the application will help to effectively manage request retries and reduce the chance of continually hitting the provisioned throughput limit.

upvoted 3 times

Certified101 2 years, 3 months ago

Selected Answer: CE

C & E correct

upvoted 4 times

tqiu654 2 years, 3 months ago

Selected Answer: BC

Based on ChatGPT: BC

upvoted 1 times

Claire_KMT 2 years, 5 months ago

B. Increase the number of read capacity units (RCUs) that are provisioned for the DynamoDB table.
OR
E. Change the capacity mode of the DynamoDB table from provisioned to on-demand.
C. Reduce the frequency of requests to DynamoDB by implementing exponential backoff.

upvoted 1 times

tapan666 2 years, 5 months ago

It 'inserts' data, so it needs WCUs and not RCUs. So option B is invalid too. C and E are the correct options.

upvoted 10 times